

Akshunya Vijayvargiya

3rd Year Undergraduate

Department Computer Science and Engineering

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📍 Akshunya1101 | in akshunya-vijayvargiya

ACADEMIC QUALIFICATIONS

Year	Degree/Certificate	Institute	CPI/%
2020 - 2024	B.Tech	Indian Institute of Technology Kanpur	9.24/10
2020	CBSE(XII)	Lord Buddha Public School, Ranpur	92.8%
2018	CBSE(X)	Atomic Energy Central School No. 4, Rawatbhata	85.6%

SCHOLASTIC ACHIEVEMENTS

- Received **Academic Excellence Award** in the academic year **2020-21** for exceptional performance in academics
- Secured **All India Rank 171** in Joint Entrance Examination (JEE) Advanced 2020 among **2.5 Lakh candidates**
- Secured **All India Rank 1291** in Joint Entrance Examination (JEE) Mains 2020 among **1.1 million candidates**
- Secured **All India Rank 257** and received a research scholarship in Kishore Vaigyanik Protsahan Yojana, 2019
- Among **National top 1%** in National Standard Examination in Physics(NSEP) and Chemistry(NSEC) 2019
- Cleared Round 1 of Summer of BitCoin, 2022 and got selected for proposal round out of **20 thousand** applicants

KEY PROJECTS

Semantic Segmentation of Legal Documents 🔄 | Ongoing

May'22 - Present

Mentor: Prof. Ashutosh Modi, CSE, IIT Kanpur

- Objective:** To develop a deep learning model for automatically segmenting legal papers via 13 rhetorical roles
- Worked on **150+** Legal Documents to automatically extract Precedents and Statutes with an accuracy of **90%**
- Performed Sentiment analysis using **nlTK** and **spaCy** libraries to classify the precedents into 4 rhetorical roles

IITKart 🔄 | CS253 Course Project, under Prof. Indranil Saha

Jan'22 - Apr'22

- Developed a selling/buying portal using frameworks like **Django** and **Bootstrap** and deployed it on **Heroku**
- Drafted software documentation including **Design document**, **Implementation document** and **User Manual**
- Executed **Unit, Integration and System Testing** of the portal with a test coverage of **80-100%** for various features

Intro to ML 🔄 | Programming Club, IIT Kanpur

May'21- Jul'21

- Exposure to data processing libraries- **Numpy**, **Pandas**, **Matplotlib** and 8 different Machine Learning models
- Applied **3 linear** and **5 non-linear** Machine Learning models from **scikit-learn** library on Flight Take Off Data
- Compared the errors using **One Hot Encoding (min. RMSE 5.739)** and **Label Encoding (min. RMSE 5.817)**

Deep Dive Into Algorithms 🔄 | Association for Computing Activities, IIT Kanpur

May'21-Jul'21

- Detailed learning of various containers and their associated functions in the **Standard Template Library of C++**
- Studied a variety of algorithms related to the topics of **Number Theory**, **Binary Search** and **Graph Theory**
- Solved various problems in codeforces contests by implementing the learned algorithms using STL containers

SAT Solver | CS202 Course Project, under Prof. Subhajit Roy

Feb'22

- Implemented a SAT Solver in **python** using **DPLL Algorithm**, involving **unit propagation** and **literal elimination**
- Improved efficiency of the algorithm to a minimum time of **35 seconds** for a CNF with **50 literals** and **218 clauses**
- Created a **k×k Sudoku Solver** in Python using **PySAT toolkit**, by finding boolean satisfiability of clauses in CNF

Library Management System 🔄 | CS253 Course Project, under Prof. Indranil Saha

Apr'22

- Built a Library Management System using **Object Oriented Programming** in **C++** to manage book transactions
- Utilized **Inheritance** concept to make a user class and 3 sub-classes - namely Student, Professor and Librarian
- Provided various functionalities to different users to search, borrow/return books or calculate and pay their fines

COMPETITIVE PROGRAMMING

- Highest rating **1615(Expert)** on Codeforces (AkshunyaVijayvargiya) and **2071(5 star)** on Codechef (akshunya1101)
- Secured common list rank **18** in Codeforces Round #799 among more than **33 thousand** participants

TECHNICAL SKILLS

Programming: C, C++, Python **Exposure:** Verilog HDL, MIPS Assembly Language, Javascript, Bash, HTML, CSS

Utilities and Frameworks: Git/GitHub, $\LaTeX$, Django, PySAT, MXNet, PyTorch, NumPy, Pandas, Matplotlib, Seaborn

RELEVANT COURSEWORK

Data Structure and Algorithms	Probability for Computer Science	Fundamentals of Computing
Software Development and Operations	Computer Architecture	Logic in Computer Science
Discrete Mathematics	Linear algebra and ODE	French - Level I

POSITIONS OF RESPONSIBILITY

Academic Department Mentor, Academics and Career Council IIT Kanpur

Jul'22 - Present

- Collaborate with Academics and Career council to guide and help a batch of 135 CSE students in their academics

Project Mentor, BlockChain101, ACA IIT Kanpur

May'22 - Jul'22

- Mentored a group of 24 students to help them learn about various components of **Blockchain Technology**
- Provided resources and assignments to the mentees and finally worked on making their Blockchain network

Secretary, Humour House IIT Kanpur

Jun'21-Apr'22

- Learning of performing skills like Stand Up comedy, Improvisation, Roast and teaching that to other enthusiasts
- Management of club events and competitions along with participating by showcasing my performance in them